

AETHER EYE - SEMANTIC SLAM AND LIDAR MAPPING WITH MLLM GUIDED NAVIGATION FOR VISUAL ASSISTANCE

A MAJOR PROJECT REPORT

Submitted by

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to

APJ Abdul Kalam Technological University

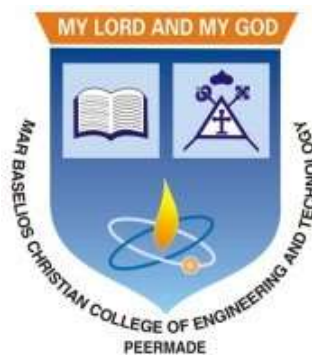
In partial fulfillment of requirements for the award of degree

of

Bachelor of Technology

In

Computer Science and Engineering



**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING MAR BASELIOS CHRISTIAN COLLEGE OF
ENGINEERING AND TECHNOLOGY, PEERMADE**

OCTOBER 2025

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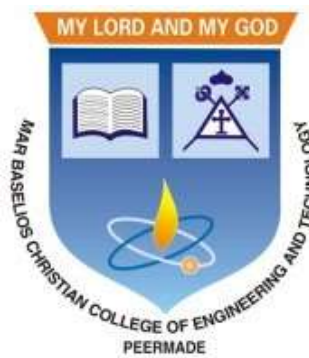
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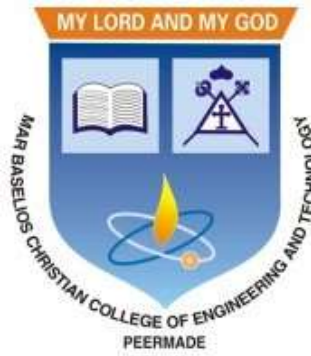
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OCTOBER 2025

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
MAR BASELIOS CHRISTIAN COLLEGE OF ENGINEERING AND TECHNOLOGY
2025-2026



CERTIFICATE

This is to certify that this report entitled “**AETHER EYE - SEMANTIC SLAM AND LIDAR MAPPING WITH MLLM GUIDED NAVIGATION FOR VISUAL ASSISTANCE**” submitted by **ADHIL BIJU(MBC22CS007), AJAS MATHEW(MBC22CS011), ALAN TAITTUS (MBC22CS014), DEVA NANDH P R(MBC22CS030) and JEWEL LEELU SUNIL (MBC22CS041)** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the degree of B.Tech. in Computer Science and Engineering, is a bonafide report of the project work carried out by them under the guidance and supervision of **Prof. Aryalakshmi R**, Assistant Professor, Department of Computer Science and Engineering. This report in any form has not been submitted to any other University or Institute for any purpose.

Prof. ARYALAKSHMI R
Project Guide
Assistant Professor
Department of CSE

Dr. Ushus Maria Joseph
Head of the Department
Department of CSE

DECLARATION

We hereby declare that this report entitled “**AETHER EYE - SEMANTIC SLAM AND LIDAR MAPPING WITH MLLM GUIDED NAVIGATION FOR VISUAL ASSISTANCE** ”is a bonafide record of the preliminary project work carried out by us under the super- vision of **Prof. ARYALAKSHMI R** in the Department of Computer Science and Engineering Mar Baselios Christian College of Engineering and Technology, Peermade. The project proposal presented in this report have been presented for the award of any other degrees.

Peermade

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ABSTRACT

Visual impairment (VI) affects a large portion of the global population, creating a need for reliable and non-intrusive electronic travel aids (ETAs) that support independent navigation. Existing aids often suffer from limited accuracy, restricted detection range, and dependence on environment-specific infrastructure. This project introduces an AI-driven assistive system powered by a Multi-Modal Large Language Model (MLLM) integrated with advanced sensor fusion, combining LiDAR and vision-based (RGB-D) data. Designed as a lightweight wearable device, the system addresses two key challenges: accurate real-time obstacle detection and recognition—including uneven surfaces and transparent objects—and intuitive, natural language-based guidance. LiDAR sensors ensure high precision and low latency (below 100 ms) for collision avoidance, while a Convolutional Neural Network (CNN)/YOLO model processes visual input for object identification and semantic mapping. The MLLM synthesizes multi-modal data into natural voice commands and haptic feedback, providing a high-range, efficient navigation solution. Experimental results demonstrate over a 200% increase in walking speed for trained users and a significant reduction in detection latency, confirming the potential of this AI-based wearable system to enhance mobility, safety, and independence for visually impaired individuals.

INSTITUTION VISION

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Our efforts are dedicated for developing a learner centric education environment to:

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- Practice real world problem solving
- Foster team work in engineering design
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DEPARTMENT MISSION

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Student of the Computer Science and Engineering program will:

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PROGRAM OUTCOMES (POs)

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- PO2:** Problem Analysis : Identify, formulate, review research literature , and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- PO3:** Design / Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
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- PO5:** Modern Tool Usage : Create , select , and apply appropriate techniques , resources , and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
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- PO7:** Environment and Sustainability : Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

- PO8:** Ethics : Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- PO9:** Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings
- PO10:** Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- PO11:** Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- PO12:** Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change

PROGRAM SPECIFIC OUTCOMES (PSOs)

- PSO1:** Apply theoretical and practical knowledge to analyse and model hardware systems with varying complexity.
- PSO2:** Utilize modern programming platforms, tools and logical reasoning skills to analyse and develop application software.
- PSO3:** Apply computational and networking skills to solve and develop secure communication environment.

COURSE OUTCOMES (COs)

CO1	Model and solve real world problems by applying knowledge across domains (Cognitive knowledge level: Apply).
CO2	Develop products, processes or technologies for sustainable and socially relevant applications (Cognitive knowledge level: Apply).
CO3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks (Cognitive knowledge level: Apply).
CO4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms (Cognitive knowledge level: Apply).
CO5	Identify technology/research gaps and propose innovative/creative solutions (Cognitive knowledge level: Analyze).
CO6	Organize and communicate technical and scientific findings effectively in written and oral forms (Cognitive knowledge level: Apply).

Mapping of course outcomes with program outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	1	2	2	2	1	1	1	1	2
CO2	2	2	2		1	3	3	1	1		1	1
CO3									3	2	2	1
CO4					2			3	2	2	3	2
CO5	2	3	3	1	2							1
CO6					2			2	2	3	1	1

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CHAPTER 1

INTRODUCTION

The growing global population affected by visual impairment (VI) necessitates the development of highly reliable and non-intrusive Electronic Travel Aids (ETAs) to support independent navigation and improve overall quality of life [13]. Over 80% of human information acquisition relies on sight, underscoring the critical nature of technologies that can effectively compensate for this deficit in complex environments [12]. While traditional aids like the white cane are fundamental, they offer only limited proximity warning and lack the contextual awareness required for safe, autonomous travel in dynamic or unfamiliar settings [15].

This project proposes a novel AI-based framework, the AI-Driven Assistive Navigation System, that employs advanced Multi-Modal Large Language Models (MLLMs) and sensor fusions specifically combining high-accuracy Lidar/Time-of-Flight (ToF) sensors with cloud-accelerated vision technology—to identify environmental elements and provide nuanced, natural language guidance in real-time [3].

1.2 The Need for MLLM based context description in visual aids

The pursuit of true independence for visually impaired (VI) individuals is constrained by several critical limitations in existing and prototype electronic travel aid (ETA) solutions, which this work aims to overcome [12]. Many advanced indoor navigation systems rely on environment-specific infrastructure such as RFID, BLE beacons, or Wi-Fi fingerprinting, making them dependent on pre-installed location tags [13][15]. This severely limits their universality and mobility, rendering them ineffective in unfamiliar or unmapped public spaces. Traditional sensors, including ultrasonic and infrared (IR) devices, provide limited detection accuracy and scope, often struggling with transparent objects like glass walls and offering a restricted operating range (typically ≤ 4 m), which hinders proactive navigation [1][3]. Moreover, conventional ETAs lack semantic context, relying on simple buzzers or beeps that only convey distance without identifying the nature of the obstacle, such as a chair, table, or stairs thereby increasing cognitive load and uncertainty for users [3][5]. Vision-based systems, while capable of providing richer context, face computational bottlenecks; deploying deep learning models like CNNs or YOLO on low-power microprocessors (e.g., Raspberry Pi) leads to GPU-intensive processing and high latency (up to ~ 15 seconds), making real-time guidance impractical. This project addresses these challenges through a Multi-Modal AI framework that integrates hardware and cloud-based optimization to deliver safe, scalable solution.

1.2 Sensor Fusion and Advanced Perception

The design is engineered on the principle of sensor fusion to build a comprehensive, ecologically valid perception model, integrating the strengths of distance sensing and contextual intelligence [3][7][13]. **Lidar (Time-of-Flight) Sensors:** These are chosen as the primary proximity sensors for their superior performance. Lidar (ToF) sensors offer high precision (centimeter-level), a quicker reading frequency compared to sound-based ultrasonic systems, and a small, lightweight form factor suitable for wearable integration. Utilizing an array of these sensors at different body levels (chest and head) allows for the automatic determination of an obstacle's height, size, and location [1][3]. **Dynamic Ground Awareness:** A specialized Lidar unit is angled downward to continuously monitor the terrain. This enables a unique capability for detecting potholes, downward steps, and slopes by tracking instantaneous deviations from a learned ground reference distance, providing proactive warning for ground hazards. **Semantic Vision and MLLM (Cognitive Layer):** The system employs a wearable Pi Camera/RGB-D sensor to capture the visual scene. This visual input is processed via the cloud using an established object detection model (e.g., YOLO V5s for high-confidence object and target detection). This approach provides the crucial semantic context (e.g., "Dining table," "Person") that simple distance sensors lack. **Multi-Modal Synthesis and Guidance:** The final output is formulated by an MLLM core that synthesizes data from all sensors—Lidar distance, visual object labels, and GPS coordinates—into coherent, actionable voice commands delivered to the user [12]. This natural language output drastically lowers the user's cognitive load and enhances safety [7].

1.3 Real-Time Safety and Semantic Awareness: A Distributed Architecture

The core strength of the system lies in its distributed, dual-path processing architecture, which is engineered to overcome the real-time constraints typical of monolithic systems [3]. This design establishes a crucial balance between rapid physical safety and intelligent semantic awareness. The Low-Latency, Localized Safety Loop utilizes dedicated microcontrollers (like the Arduino Nano) linked directly to each Lidar and vibrating motor. This hardware-accelerated pathway executes simple, time-critical logic (e.g. $\text{Distance} < 30 \text{ cm}$) to ensure the quickest possible response for imminent collision warnings, striving for the critical target of 100 ms latency [7][8]. Concurrently, the High-Intelligence, Cloud-Accelerated Loop strategically bypasses the slow local processing of the Raspberry Pi by outsourcing GPU-intensive visual and semantic tasks (MLLM) to a Cloud Vision API (e.g., Google Vision API). This strategic division allows the system to deliver rich, contextual information (e.g., confirming "cash desk" or "bus stop" signs) within a sufficient timeframe of approximately 3 seconds for natural language guidance. This dual-priority strategy thus technical superiority, providing robust and highly efficient navigation for the visually impaired [13].

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

The shift towards creating automated assistive technologies for the visually impaired (VI) represents a convergence of multiple advanced engineering disciplines, spanning computer vision, robotics, and human-computer interaction [12]. This literature review is strategically designed to establish the historical context and evaluate the current state-of-the-art across key dimensions relevant to developing the next generation of Electronic Travel Aids (ETAs) [13]. Previous ETA solutions have been fundamentally limited by reliance on external infrastructure, insufficient detection range, and computational latency, all of which hinder the VI user's ability to achieve genuine independent mobility. This survey critically analyzes modern advancements in sensor fusion, spatial awareness, and semantic interpretation to justify the proposed system's core architecture [12][13]. The research reviewed is organized to reveal the persistent gaps in providing contextual, low-latency, and universally effective guidance, thereby building a strong foundation for the proposed AI-Driven Assistive Navigation System[12].

2.2 INTELLIGENT HEAD-MOUNTED OBSTACLE AVOIDANCE WEARABLE FOR THE BLIND AND VISUALLY IMPAIRED

This study details the development and evaluation of an intelligent head-mount device using an array of ultrasonic sensors and a 9-DOF Inertial Measurement Unit (IMU) to provide obstacle avoidance warnings in real-time. The primary goal was to create a computationally efficient and compact mechanism using a resource-limited device, such as a Raspberry Pi 4B, for reliable, real-time detection [1]. The system focuses on hardware and software integration and utilizes a multithreaded parallel processing approach to manage multiple sensor inputs simultaneously, which effectively avoids task blocks and accelerates system efficiency on limited resources. The study experimented with various machine learning models (including decision trees, neural networks, and ensemble methods) and concluded that the Decision Tree was the most suitable algorithm, offering the optimal balance of accuracy and speed for a lightweight solution. The system was tested in a controlled indoor environment, accounting for natural head turns and sensor positioning variations [1]

ADVANTAGES

Our solution employs Lidar (ToF), which is superior to the system's reliance on ultrasonic sensors that offer low frequency readings, lack accuracy, and only provide non-semantic distance. We integrate the complexity of vision via a Cloud MLLM to provide rich natural language scene descriptions, far exceeding the system's simple audio beep warning. Our design incorporates features like pothole detection and GPS guidance, extending functionality beyond the reactive obstacle avoidance focus of this device.

DISADVANTAGES

The reliance on ultrasonic sensors for distance measurement provides limited spatial fidelity and reduced reliability compared to LiDAR technology. Ultrasonic sensors often suffer from beam divergence, low resolution, and sensitivity to environmental factors such as temperature and surface texture, leading to inconsistent range estimations. Moreover, the exclusion of computer vision restricts the system's ability to interpret surroundings, reducing its output to a simple, non-semantic alert (a prolonged beep) that signals obstacles without context. This lack of semantic understanding prevents users from making informed navigation decisions and compromises safety. The system's dependence on direct-path detection—where head movements are avoided to prevent interference—further limits spatial awareness, leaving side and peripheral obstacles undetected. These shortcomings highlight the need for a more advanced, context-aware solution that integrates LiDAR precision with computer vision and AI-driven semantic interpretation for richer and more reliable navigation assistance.

2.3 A DESIGN, DEVELOPMENT AND PERFORMANCE ANALYSIS OF COGNITIVE ASSISTING AID WITH MULTI SENSOR FUSED NAVIGATION FOR VISUALLY IMPAIRED PEOPLE

This prototype, the "Blind's Apron," utilized a Multi-Sensor Fused Navigation approach, replacing ultrasonic sensors with VL53L1X Time-of-Flight (ToF) sensors for enhanced accuracy and speed. It employs multiple sensors connected to individual Arduino Nanos for rapid haptic output (vibrating motors) upon detecting obstacles within 30 cm. A specialized, downward-facing sensor detects potholes and uneven surfaces. Crucially, semantic object detection is achieved by outsourcing processing to the Google Cloud Vision API via the Raspberry Pi. User trials demonstrated a dramatic improvement in trained user walking speed of over 200% [3].

ADVANTAGES

Our solution is built upon and refines this foundation, focusing on improving the hardware form factor from an "apron" to a compact cap. We aim to drastically reduce the critical latency, which averaged 710 ms in this system, by optimizing the processing path and utilizing ROS 2 industrial standards for synchronization. Our GPS function will provide proactive, turn-by-turn guidance to the user, contrasting with this system's use of GPS solely for caretaker tracking [3].

DISADVANTAGES

The most significant constraint is the high total system latency of 710 ms, requiring the user's walking speed to be strictly limited to 3km/h (to prevent collisions [3]). The device is categorized as a high-power consumption system, with continuous operation lasting only 3–5 hours. Furthermore, the entire object detection feature relies on continuous connectivity to the Cloud Vision API, risking failure in areas with limited or no internet access. Finally, the system requires a 4 to 5 second initial delay upon startup to configure the ground reference for pothole detection [3].

2.4 A WEARABLE VISUALLY IMPAIRED ASSISTIVE SYSTEM BASED ON SEMANTIC VISION SLAM FOR GRASPING OPERATION

This study presents a wearable assistive system designed to aid visually impaired users in grasping nearby objects within a limited tabletop environment. The system uses Semantic Visual SLAM, combining the structural mapping capabilities of the ORB-SLAM2 algorithm with the classification accuracy of a YOLO-based model to create a detailed three-dimensional semantic map [2]. A glove equipped with a depth camera captures data, while a vibration array provides tactile feedback to indicate object distance and direction. The system demonstrated high positional accuracy for close-range object manipulation [2].

ADVANTAGES

Our system extends this concept beyond confined environments by integrating LiDAR and GPS to support full-scale mobility in both indoor and outdoor spaces [3]. The inclusion of a Multi-Modal Large Language Model allows for natural language-based guidance, offering a richer and more intuitive experience than simple haptic feedback. Furthermore, our design removes the dependency on an external computer, achieving a self-contained, mobile, and wearable solution [7].

DISADVANTAGES

This system is limited to tabletop object interaction rather than general navigation, restricting its practical usability for independent mobility. Its hardware requires a powerful external computer with a high-end graphics processor to handle SLAM and object detection, making it impractical as a portable device. Additionally, it demands extremely slow hand movements to maintain stable camera tracking, which would be infeasible for real-time pedestrian navigation [2].

2.5 OBSTACLE AVOIDANCE FOR BLIND PEOPLE USING YOLO ALGORITHM, DARKNET, AND GTTS

This study presents an obstacle avoidance system that integrates the YOLO v3 model with the Darknet framework for real-time object detection. The model is trained on a large, multi-class dataset containing eighty object categories and utilizes the Google Text-to-Speech (GTTS) API to deliver verbal cues describing detected obstacles and their relative locations. The primary goal of this work is to enhance obstacle avoidance performance compared to traditional sensor-based systems by leveraging visual recognition and immediate auditory feedback [4].

ADVANTAGES

Our proposed system builds upon this foundation by incorporating multimodal sensor fusion that combines LiDAR and vision data to achieve far greater distance accuracy and depth perception than this camera-only approach.[4].The integration of a Multi-Modal Large Language Model (MLLM) allows for the generation of complex, context-aware natural language scene descriptions that extend beyond the basic verbal alerts produced through GTTS. Additionally, our design integrates GPS and semantic mapping for proactive route planning and guidance, transforming navigation from a purely reactive process into a predictive, intelligent system [5].

DISADVANTAGES

The referenced solution depends solely on a single camera for environmental input and lacks additional sensors necessary for accurate range and distance measurement. Its vision-only architecture restricts it to reactive detection , providing alerts only after obstacles enter the field of view, whereas our system anticipates potential hazards through predictive mapping. Furthermore, its recognition capabilities are limited to the eighty predefined object classes within its training dataset and cannot describe novel or complex environmental scenarios, reducing its effectiveness in unstructured or dynamic real-world settings [4].

2.6 YOLOV3-BASED SMART ASSISTIVE NAVIGATION SYSTEM FOR VISUALLY IMPAIRED INDIVIDUALS

This study presents a wearable assistive prototype that integrates a Raspberry Pi 4 processing core, a Pi Camera, an ultrasonic sensor, and a GPS module for real-time navigation support [5]. The system employs the YOLOv3 object detection model to recognize obstacles and provides audio-based feedback through voice assistance, enabling users to receive immediate guidance during movement. By combining vision, proximity, and location data, the solution aims to enhance mobility and independence for visually impaired individuals in both indoor and outdoor environments [5].

ADVANTAGES

Our proposed model advances this approach by replacing ultrasonic sensors with high-accuracy LiDAR technology to address the low fidelity and limited three-dimensional mapping capabilities of ultrasonic designs [4]. We also substitute the basic GPS module with a higher-precision, semantically aware positioning system that integrates environmental understanding, thereby improving accuracy and reliability under diverse conditions [5]. Furthermore, our cloud-accelerated Multi-Modal Large Language Model (MLLM) framework mitigates latency and processing overhead, delivering faster, context-rich navigation instructions that extend beyond the reactive capabilities of the referenced system [7].

DISADVANTAGES

The reliance on ultrasonic sensors limits the system's spatial accuracy and depth perception compared to LiDAR-based approaches. The positional reliability of the standard GPS module is prone to degradation in densely built or obstructed areas, which poses a risk to navigation accuracy. Additionally, the camera module's limited resolution and the uneven weight distribution of the wearable hardware reduce overall user comfort and operational efficiency [5].

2.7 SMART CAP FOR VISUALLY IMPAIRED PERSONS USING RASPBERRY PI

This paper introduces a low-cost, head-mounted “Smart Cap” designed for obstacle detection and basic environmental awareness. The system employs a Raspberry Pi 3, an ultrasonic sensor, and a standard web camera. Visual data are processed using image processing techniques implemented through the OpenCV library, while auditory feedback is provided through a text-to-speech synthesizer [7]. The design’s primary objective is to create an affordable and portable safety device suitable for visually impaired users, particularly in emergency or disaster scenarios [7].

ADVANTAGES

Our system utilizes LiDAR sensors to deliver precise distance measurement, overcoming the accuracy limitations inherent in ultrasonic technology. We enhance visual analysis through the integration of advanced deep learning techniques, including YOLO and MLLM frameworks, enabling robust object classification and scene understanding that far surpass traditional OpenCV-based image processing [7]. Furthermore, our model introduces long-range navigation through GPS and simultaneous localization and mapping (SLAM), transforming the system from a purely reactive aid into a proactive, context-aware navigation assistant [5].

DISADVANTAGES

The “Smart Cap” design’s reliance on ultrasonic sensing restricts the precision and depth of obstacle detection. Its simple vision model provides only basic alerts, such as beeps or text-to-speech messages, lacking semantic depth or natural language comprehension. Although capable of limited facial detection, the use of older image processing methods prevents generalization across diverse environments, reducing its adaptability and effectiveness compared to contemporary AI-driven solutions [7].

2.8 SMART ASSISTIVE NAVIGATION SYSTEM FOR VISUALLY IMPAIRED PEOPLE

This research introduces a comprehensive assistive navigation framework that combines a smart stick with a mobile phone application to enhance the independence and safety of visually impaired users in both indoor and outdoor settings[8]. The system employs a Raspberry Pi 4 for data fusion and sensor integration, achieving real-time object detection using the YOLOv8 model. A sensor array consisting of ultrasonic, moisture, and camera modules supports environmental perception, while guidance is delivered through a multimodal interface that combines voice-based auditory cues with vibration-based tactile alerts[8]. The accompanying Android application provides GPS-based route assistance and voice-command functionality, allowing seamless interaction between the user and the system.

ADVANTAGES

Our proposed model utilizes LiDAR-based ranging for high-precision spatial measurement, addressing the limited depth fidelity and low accuracy of ultrasonic sensors used in the referenced system. Designed for energy efficiency and lightweight operation, our framework distributes processing workloads across microcontrollers, improving portability and power management. Additionally, through cloud-accelerated computation and optimized AI pipelines, we achieve significantly reduced detection latency and enhanced real-time responsiveness, providing a more reliable and collision-free navigation experience [8][11].

DISADVANTAGES

The studied system relies on simple ultrasonic sensors for basic depth estimation, which restricts its ability to construct detailed three-dimensional environmental maps. Its high power consumption and bulky hardware configuration limit both portability and continuous operation. Furthermore, the model demonstrates classification inconsistencies when distinguishing between visually similar objects and struggles to generalize effectively in unstructured or unfamiliar environments, reducing its robustness for real-world mobility applications [8].

2.9 VOICE-BASED SMART NAVIGATION SYSTEM FOR BLIND PEOPLE

This project introduces a voice-guided smart wearable aid designed for visually impaired individuals, mounted on the user's shoe for unobtrusive use. The system is powered by a Raspberry Pi Zero processor and incorporates a GPS module alongside dual ultrasonic sensors. It provides real-time obstacle detection, human recognition, and navigation assistance through spoken alerts delivered via earphones or a microphone. The design prioritizes cost-effectiveness and mobility while identifying power supply management as a critical constraint. Navigation is achieved through the Mapbox API, which provides route planning and directional guidance based on the user's destination.

ADVANTAGES

Our design enhances this concept by incorporating LiDAR (Time-of-Flight) sensors, which deliver higher accuracy and faster distance readings compared with the ultrasonic sensors used in this model. By integrating a Multi-Modal Large Language Model (MLLM), our system provides richer, more contextual scene descriptions rather than basic proximity or directional alerts. Furthermore, we utilize optimized power architectures and more capable processors to overcome the energy and performance limitations of the Raspberry Pi Zero, ensuring efficient real-time operation [10].

DISADVANTAGES

The dependence on ultrasonic sensors restricts range and accuracy, providing only short-distance detection compared with more robust LiDAR-based solutions. As the device is shoe-mounted, it is inherently limited to detecting ground-level obstacles, critically failing to sense aerial or upper-body obstructions such as low-hanging tree branches, open cabinet doors, or elevated signage. Additionally, the system lacks semantic or visual context, offering only basic proximity alerts without meaningful environmental understanding. Consequently, the user receives an alert for an object but cannot distinguish a harmless curb from a dangerous drop-off, or a temporary construction cone from a permanent structural column. This deficiency in object identification and context forces the user to rely heavily on other senses and slows their decision-making process, ultimately degrading the speed and confidence of their navigation.

2.10 SMART CAP - WEARABLE VISUAL GUIDANCE SYSTEM FOR BLIND USERS

This research presents an affordable head-mounted “Smart Cap” system designed to assist visually impaired individuals with object recognition and navigation. The setup includes a NoIR camera connected to a Raspberry Pi 3 processor. It utilizes TensorFlow and a pre-trained Convolutional Neural Network (CNN) based on the SSD MobileNet architecture to detect and classify up to ninety object types from the COCO dataset [9]. The system converts detection outputs into audible speech using the eSpeak Text-to-Speech synthesizer, providing the user with basic object identification and situational awareness [9].

ADVANTAGES

Our system advances this concept through LiDAR and sensor fusion, providing accurate depth and distance perception that overcomes the limitations of a vision-only model. By offloading deep learning computations to the cloud and integrating an MLLM, our design eliminates the high latency caused by GPU-intensive processing on the Raspberry Pi 3 [9]. Additionally, we incorporate proactive path-planning capabilities through GPS and SLAM technologies, transforming simple obstacle detection into a comprehensive, context-aware navigation framework [5][7].

DISADVANTAGES

The use of a single camera and locally executed deep learning leads to high computational demand and latency. Object recognition remains limited to the ninety predefined classes from the COCO model, restricting adaptability to complex real-world environments [9]. Moreover, the reliance on the SSD-MobileNet model prioritizes speed over accuracy, reducing detection precision, especially in challenging lighting or cluttered conditions [9].

2.11 MULTI-AGENT DEEP REINFORCEMENT LEARNING FOR MULTI-ROBOT APPLICATIONS: A SURVEY

This paper provides an extensive survey of Multi-Agent Deep Reinforcement Learning (MADRL) in multi-robot systems, exploring its applications in areas such as path planning, coordinated object transport, and swarm behavior. It discusses fundamental concepts including Deep Reinforcement Learning (DRL), Markov Decision Processes (MDP), and learning frameworks like DQN and PPO [10]. The study emphasizes that LiDAR and multi-spectral cameras are essential for enabling autonomous coordination and perception in multi-robot environments [10]. The authors identify scalability and the gap between simulation and real-world implementation as key challenges in applying MADRL effectively.

ADVANTAGES

Although our project focuses on a single-user assistive device rather than multi-agent robotics, this study reinforces the value of LiDAR and advanced imaging for multi-modal perception. It also supports the use of sophisticated AI models such as MLLMs and reinforcement learning for adaptive decision-making. The discussion of transferability challenges justifies our choice of cloud-based processing to bridge the gap between simulated optimization and real-world adaptability[10].

DISADVANTAGES

The scope of this survey is centered on multi-agent robotic systems rather than human-wearable assistive technologies, making direct translation to our design context limited. Additionally, MADRL algorithms introduce unnecessary computational complexity and power demands for a single-agent wearable system primarily intended for localized environmental awareness and navigation [10].

2.12 RESEARCH AND IMPLEMENTATION OF TRAVEL AIDS FOR BLIND AND VISUALLY IMPAIRED PEOPLE

Our proposed framework builds upon this concept while introducing several significant advancements. Whereas this system utilizes a purely vision-based model on the Jetson Nano, our architecture incorporates LiDAR fusion, offering greater precision and modality-independent distance measurement. Furthermore, we integrate a Multi-Modal Large Language Model (MLLM) that delivers detailed natural language descriptions, representing a substantial improvement over the vibration-based feedback employed here. The demonstrated success of the Jetson Nano platform in this study supports our selection of similar edge-computing hardware for localized processing within our design[11].

ADVANTAGES

Our proposed framework builds upon this concept while introducing several significant advancements. Whereas this system utilizes a purely vision-based model on the Jetson Nano, our architecture incorporates LiDAR fusion, offering greater precision and modality-independent distance measurement. Furthermore, we integrate a Multi-Modal Large Language Model (MLLM) that delivers detailed natural language descriptions, representing a substantial improvement over the vibration-based feedback employed here[11]. The demonstrated success of the Jetson Nano platform in this study supports our selection of similar edge-computing hardware for localized processing within our design [6].

DISADVANTAGES

Although the vision-only approach effectively reduces hardware costs, it sacrifices the redundancy and precision provided by multi-sensor fusion with LiDAR [11]. The vibration feedback mechanism, while simple and low-power, is less effective in outdoor conditions compared with voice-based or multimodal auditory cues. Additionally, the system's processing remains limited to on-device computation without leveraging cloud-based resources, restricting its ability to perform complex contextual scene interpretation or integrate with broader data-driven guidance frameworks.

2.13 ARTIFICIAL INTELLIGENCE FOR VISUALLY IMPAIRED INDIVIDUALS

This review explores the application of artificial intelligence in two key domains for visually impaired individuals: medical image diagnosis and assistive technologies. It discusses the use of deep learning models—such as convolutional neural networks (CNNs), generative adversarial networks (GANs), and transfer learning—for diagnosing conditions like diabetic retinopathy and glaucoma from retinal images. In the context of assistive devices, the paper highlights how AI has been integrated into smart canes, glasses, and mobile systems to assist with obstacle detection, text reading, and even artistic or creative tasks [12]. However, most existing aids analyzed in this review are limited in spatial awareness, capable of detecting only nearby obstacles, and often lack full-scene understanding and contextual interpretation [12].

ADVANTAGES

Our system extends beyond the scope of existing AI-based visual aids by combining multimodal sensor data (LiDAR, RGB-D, and visual inputs) to deliver full 360-degree environmental awareness, surpassing the narrow-range detection capabilities of the systems described in the review. We also integrate a Multi-Modal Large Language Model (MLLM) to transform raw sensor data into natural language guidance and semantic scene descriptions, significantly enhancing interpretability and user experience [10]. Unlike many existing systems that focus on single-function applications—such as text reading or object avoidance—our framework unifies OCR, obstacle detection, and dynamic route planning into one cohesive, proactive navigation assistant [10].

DISADVANTAGES

The systems discussed in this review generally prioritize deep learning for specific tasks but do not achieve the level of cross-modal integration seen in our approach. Most remain reliant on camera-only sensing, which introduces limitations under low-light or occluded conditions and increases dependency on visual clarity. Additionally, while our system leverages distributed cloud processing for complex computation, the solutions reviewed rely solely on embedded hardware, constraining real-time performance. Their cost, bulkiness, and maintenance challenges further limit practicality compared to our portable, power-efficient, and cloud-assisted architecture [12].

2.14 A COMPREHENSIVE REVIEW OF NAVIGATION SYSTEMS FOR VISUALLY IMPAIRED INDIVIDUALS

This paper provides an extensive review of navigation technologies developed for visually impaired users, categorizing solutions into four major domains: visual imagery systems, non-visual data systems, map-based navigation, and three-dimensional audio guidance. The authors highlight the growing convergence of sensors such as ultrasonic modules and LiDAR with Artificial Intelligence for real-time obstacle detection [13]. The review further emphasizes the pivotal contribution of smartphone-based solutions in enhancing accessibility and independence. It underscores the necessity of Human-Centered Design (HCD) principles to ensure user comfort, adaptability, and long-term adoption. Additionally, it discusses the advantages of integrating RGB-D cameras with tactile sensors to improve environmental perception, while acknowledging the practical challenges of data fusion, synchronization, and hardware cost [13].

ADVANTAGES

Our solution directly aligns with the insights provided in this review by combining LiDAR and RGB-D vision sensing in a unified multimodal framework, ensuring precise and consistent depth perception under diverse lighting and environmental conditions [13]. We extend the reviewed concepts by integrating AI-driven semantic understanding through an MLLM, enabling descriptive and context-aware guidance rather than simple alert-based feedback. Furthermore, our design prioritizes user-centered ergonomics lightweight, wearable form factors, and intuitive voice-haptic feedback—to ensure sustained usability and accessibility in real-world conditions, effectively operationalizing the review’s HCD recommendations [10].

DISADVANTAGES

Compared with our system, this paper-based framework provides only conceptual insights without an implemented prototype, leaving performance metrics such as accuracy, latency, and user experience untested. The reviewed systems largely rely on conventional sensor fusion methods and local computation, lacking the distributed cloud-based optimization that our model employs to reduce hardware complexity and enhance scalability. Additionally, while the review acknowledges the value of tactile sensing, it does not explore advanced multimodal feedback or real-time adaptive processing as featured in our proposed design.

2.15 AN AUTOMATED NAVIGATION SYSTEM FOR BLIND PEOPLE

This study proposes an advanced navigation system designed to support visually impaired users through a combination of vision, sensor fusion, and cloud computing. Built upon a Raspberry Pi 4 platform, the system incorporates ultrasonic sensors for distance measurement and stereo cameras for real-time video capture and depth estimation [14]. The architecture leverages deep learning techniques, such as YOLO-based object and currency recognition, while integrating additional intelligent features like facial recognition for identifying known individuals, Optical Character Recognition (OCR) for reading printed and handwritten text in English and Bengali, and speech recognition for interactive voice commands. A connected smartphone hotspot enables cloud-based processing and data storage, which supports caretaker tracking through a web or Android application [14].

ADVANTAGES

Our system extends this approach by replacing the low-accuracy ultrasonic sensors with LiDAR technology, which provides higher precision, broader range, and faster depth measurement. We incorporate Semantic SLAM and dedicated pothole detection, significantly expanding the system's environmental understanding beyond simple obstacle identification. Through the integration of a Multi-Modal Large Language Model (MLLM), our framework delivers more natural and contextually rich voice-based guidance, effectively synthesizing multimodal inputs into coherent and descriptive feedback for the user[14].

DISADVANTAGES

The reliance on ultrasonic sensors limits both the range and frequency of distance measurements, reducing accuracy compared to LiDAR-based sensing. The system's use of geometric camera-based distance estimation can produce inconsistent results under varying lighting conditions or when subjects are in motion. Additionally, performing all processing tasks—including object detection, OCR, and facial recognition—on the Raspberry Pi introduces latency and computational strain, which can hinder the system's responsiveness in dynamic, real-world environment [5][7].

2.16 INDOOR NAVIGATION SYSTEMS FOR VISUALLY IMPAIRED INDIVIDUALS

This study focuses on the development of indoor navigation systems tailored for visually impaired users, addressing the challenges of localization, path planning, and obstacle detection in enclosed or GPS-denied environments [15]. The reviewed work emphasizes the use of sensors such as ultrasonic modules, Bluetooth Low Energy (BLE) beacons, infrared, and computer vision for environmental mapping and user positioning. The integration of these technologies aims to deliver safe and independent navigation by guiding users through auditory or haptic feedback. However, the dependence on environment-specific infrastructure (e.g., pre-installed beacons or tags) limits system flexibility and scalability across different indoor settings [15].

ADVANTAGES

Our proposed system expands on the principles of this study by incorporating LiDAR and RGB-D vision sensors for dynamic 3D spatial awareness without relying on pre-installed indoor infrastructure. The integration of a Multi-Modal Large Language Model (MLLM) enables contextual natural language scene descriptions and proactive guidance, surpassing the simple direction or distance cues provided in traditional indoor navigation systems. Moreover, our framework supports seamless transition between indoor and outdoor environments using SLAM and GPS fusion, ensuring uninterrupted assistance [15]. By eliminating the need for location markers, our design promotes universality, portability, and scalability.

DISADVANTAGES

The indoor navigation approach described in this study is constrained by its reliance on static infrastructure such as BLE beacons and RFID tags, making deployment costly and maintenance-intensive. Its performance heavily depends on the density and calibration of these markers, leading to reduced reliability in dynamic or unfamiliar indoor environments. Additionally, the system typically offers limited semantic understanding—providing direction or obstacle distance but not context about object type or environmental layout. Compared with our MLLM-based framework, it lacks multimodal integration, cloud optimization, and flexible adaptability to diverse architectural conditions [15].

CHAPTER 3

PROPOSED WORK

3.1 PROBLEM STATEMENT

Existing assistive navigation technologies for the visually impaired face critical limitations that prevent users from achieving safe and unconstrained independence in dynamic, real-world environments [13][15]. The goal of this project is to solve these pervasive issues by designing a fully integrated, multimodal solution[12].

The main problem's of existing solution are mentioned here -

Many cost-effective assistive systems depend on ultrasonic sensors, which provide low spatial resolution, limited detection range (typically less than four meters), and relatively slow update rates. Documented studies indicate that system response times often average around seven hundred milliseconds, which significantly restricts user mobility by requiring very slow walking speeds to avoid potential collisions [3].

Systems that rely solely on simple sensors or basic audio feedback typically convey only distance information, such as beeps or vibrations, without describing the nature of the detected obstacle. As a result, users are unaware of whether they are approaching a chair, a staircase, or a crosswalk. This lack of semantic understanding prevents proactive decision-making and increases cognitive effort during navigation [12].

Many indoor navigation systems depend on pre-installed infrastructure such as RFID tags or Bluetooth beacons, which limits their effectiveness in new or unmarked environments. Conversely, systems that incorporate computer vision often experience severe latency due to the computational demands of deep-learning models running on low-power embedded processors [5][9].

To address these limitations, this project introduces an AI-driven, multi-modal fusion system that integrates low-latency LiDAR and Time-of-Flight sensing with cloud-accelerated vision processing through a Multi-Modal Large Language Model (MLLM). This approach enables real-time obstacle detection, semantic understanding, and comprehensive navigational guidance, offering a safer and more intelligent mobility solution for visually impaired users.

3.2 PROPOSED METHODOLOGY

The proposed system employs an asymmetric sensor fusion strategy designed to balance low-latency safety with high-level intelligence. In the Safety Tier, managed by an Arduino Mega, data from LiDAR and ultrasonic sensors are fused to enable rapid obstacle detection and distance estimation. The microcontroller executes pre-trained threshold algorithms and triggers voice feedback within 250 ms of obstacle detection, ensuring immediate user response. The Intelligence Tier, powered by a Raspberry Pi 4, handles advanced vision-based analysis through a Pi Camera and a fine-tuned Vision-Language Model (MLLM) [1]. This model generates semantic scene descriptions, recognizes familiar faces or objects, and delivers contextual insights via a Text-to-Speech (TTS) system [4][12]. The Communication and Navigation Tier integrates GPS and IMU sensor data to support outdoor navigation and fall detection [14], while dynamically switching between SLAM-based indoor positioning and GPS-based outdoor tracking to maintain continuous spatial awareness. This hierarchical architecture allows safety-critical functions to operate independently from computationally intensive visual processing enabling a unique combination of responsiveness, contextual intelligence, and wearable practicality [3].

3.2.1 CORE SYSTEM GOALS AND PERFORMANCE OBJECTIVES

The success of the proposed system is quantitatively defined by specific performance targets, validated against the dedicated test set:

1. Achieve an end-to-end response time of less than 250 milliseconds, ensuring immediate haptic and auditory feedback for enhanced user safety [3].
2. Maintain reliable detection within 10 meters using LiDAR, providing proactive hazard awareness and sufficient time for the user to react safely in both indoor and outdoor environments.
3. Enable safe and stable navigation at walking speeds of up to 1.2 meters per second without performance degradation or latency accumulation.
4. Achieve a minimum of 92% obstacle and object recognition accuracy through advanced sensor fusion and AI-based processing.
5. Deliver a minimum continuous operating time of five hours, ensuring reliable use throughout the day without frequent recharging.
6. Reach a BLEU score of at least 0.75 for the language model's descriptive output, guaranteeing natural, coherent, and contextually relevant speech narration.
7. Google TTS / eSpeak: Converts textual scene descriptions into clear audio feedback, providing the user with continuous verbal guidance [12].

8. Maintain consistent performance under varying lighting and environmental conditions, ensuring both indoor and outdoor operational reliability

3.3 REQUIREMENT SPECIFICATION

3.3.1 FUNCTIONAL REQUIREMENTS

The *Aether Eye* system performs essential functions to ensure accurate navigation assistance and situational awareness for visually impaired users.

- Object and Obstacle Detection: Detect obstacles within a 10 m range using LiDAR and provide real-time feedback.
- Scene Analysis and Description: Process camera input to generate natural-language descriptions of the surroundings [12].
- Audio Feedback: Convert detected scene information into voice output via a Text-to-Speech (TTS) engine.
- Indoor/Outdoor Navigation: Switch between SLAM-based indoor mapping and GPS-based outdoor guidance [15].
- Fall Detection: Use the IMU sensor to detect falls and issue emergency alerts.
- Sensor Fusion: Combine LiDAR, camera, IMU, and GPS data for accurate contextual understanding.
- Power Management: Monitor and optimize power usage to extend operating time.

3.3.2 SOFTWARE REQUIREMENTS

- Operating System – Raspberry Pi OS (64-bit): Provides the core environment for system computation, networking, and peripheral integration.
- OpenCV: Enables image acquisition, object detection, and frame processing, forming the visual foundation of the system.
- TensorFlow Lite / PyTorch: Executes deep-learning models efficiently on edge hardware, supporting real-time object recognition and semantic understanding.
- ROS (Robot Operating System): Coordinates SLAM, sensor fusion, and inter-module communication, ensuring smooth data flow across components.
- LangChain: Integrates language model pipelines to generate contextual and meaningful verbal responses from visual inputs.
- Google TTS / eSpeak: Converts textual scene descriptions into clear audio feedback, providing the user with continuous verbal guidance.
- NumPy / Pandas: Handle numerical computations and structured data management, supporting efficient processing and analysis.

3.3.3 HARDWARE REQUIREMENTS

- Raspberry Pi 4 Model B (4GB/8GB): Acts as the central processing unit responsible for running SLAM algorithms, AI inference, and system communication, enabling smooth and efficient data handling.
- LiDAR Sensor: Performs environmental scanning and detects obstacles within a 10 m range, providing accurate depth information for mapping and navigation [3].
- Pi Camera Module / NoIR Camera: Captures real-time video frames for visual object recognition and scene understanding, supporting AI-based perception [15].
- MPU6050 IMU Sensor: Monitors user movement, tilt, and falls, enhancing safety through continuous motion tracking and orientation detection.
- GPS Module (Neo-6M): Provides precise outdoor positioning and navigation data, supporting route tracking and guidance.
- Battery Pack: Supplies stable power for continuous operation, ensuring the system remains portable and reliable during extended use.

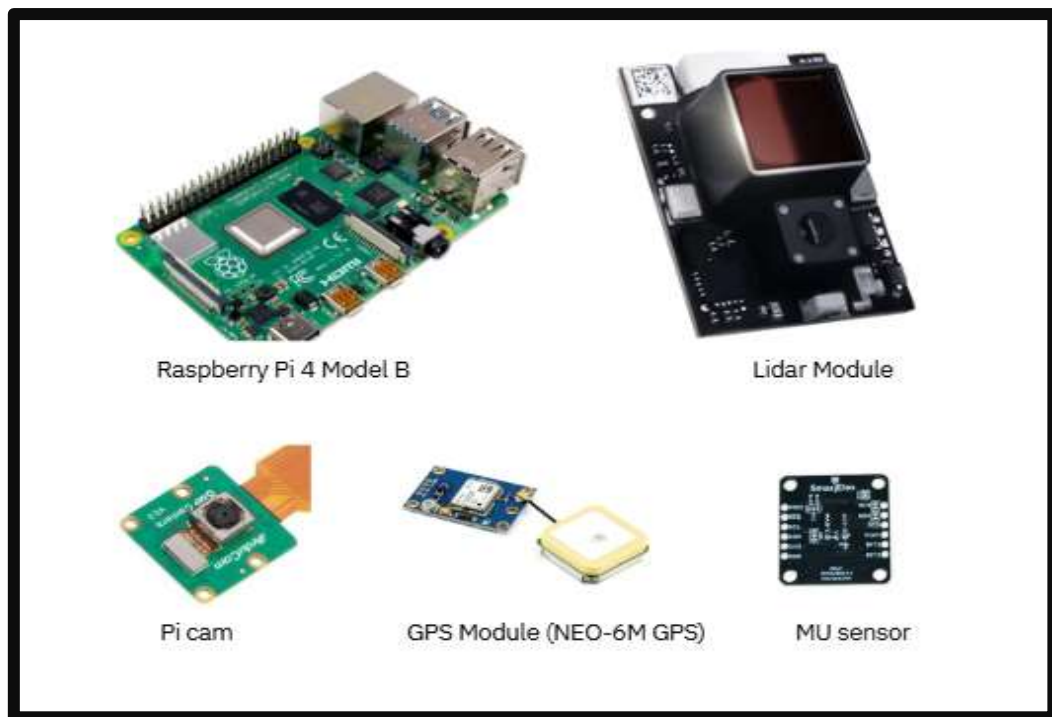


Fig 3.3.3 Hardware Components

3.4 PROPOSED WORKFLOW

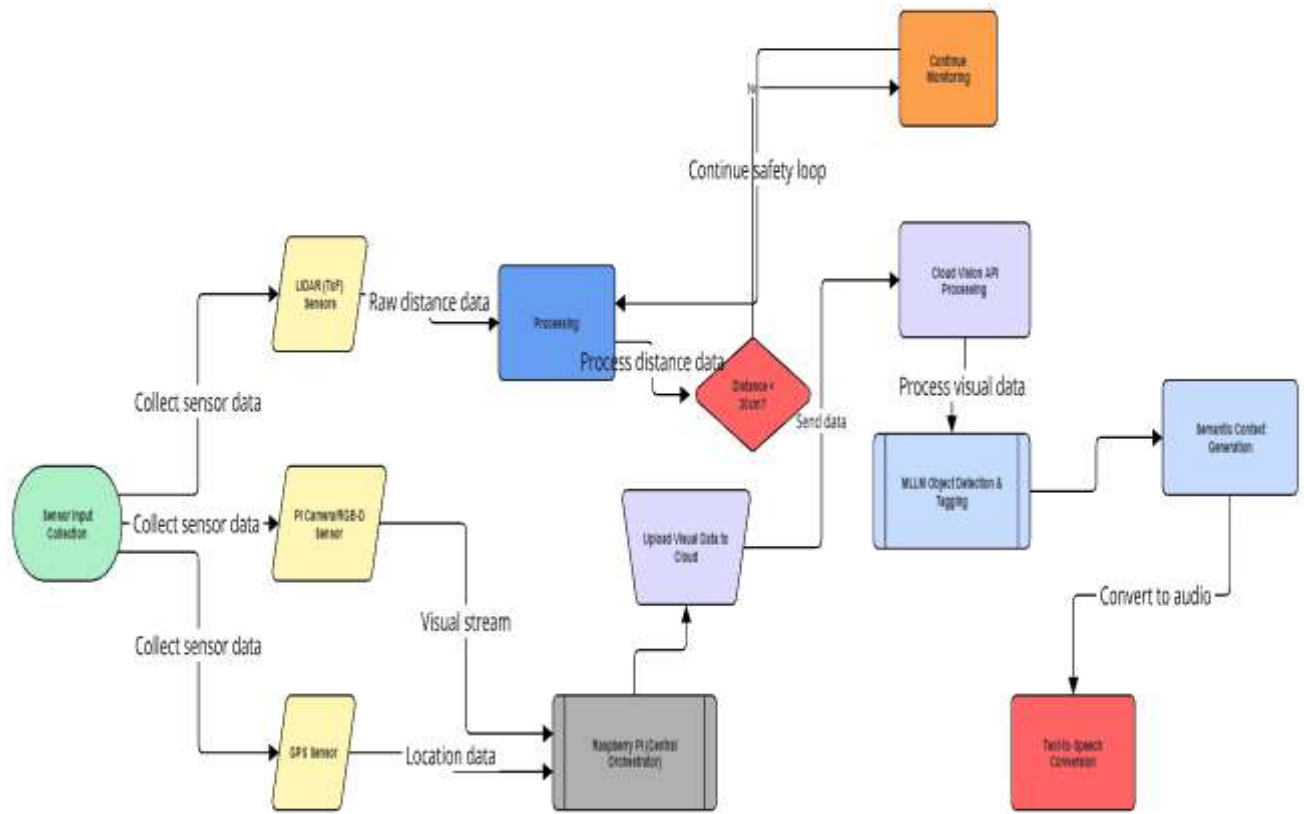


Fig 3.4 Workflow Flow Chart

The flowchart visually outlines the asymmetric processing strategy crucial for user safety, separating the immediate, low-latency sensory input (Lidar/Haptics) from the slower, resource-intensive cognitive functions (Vision/MLLM/Audio). The central element is the Raspberry Pi orchestrating the cloud-based AI to deliver complex, semantic guidance, while offloading critical collision avoidance directly to the microcontrollers. This dual-path system minimizes latency for physical safety while ensuring the user receives rich, contextual information for proactive navigation.

3.5 SYSTEM DESIGN

The Aether Eye system adopts a modular architecture that integrates sensing, computation, and feedback components into a unified wearable device [3][13]. This architecture ensures efficient data flow between perception, processing, and output layers, enabling real-time environmental awareness and mobility assistance for visually impaired users [12][12]. Each subsystem operates cohesively to provide accurate obstacle detection, intelligent scene interpretation, and seamless user communication [3][12].

3.5.1 Input Module

The Input Module forms the perception layer of the system and is responsible for capturing environmental and positional data[3][13]. It includes the LiDAR sensor, camera, IMU, and GPS module:

- LiDAR Sensor: Performs 360° environmental scanning and 10-meter range detection, enabling precise mapping and obstacle localization [3].
- Pi Camera Module / NoIR Camera: Captures continuous visual frames for real-time object recognition and scene understanding [4][14].
- MPU6050 IMU Sensor: Monitors orientation, motion, and falls to enhance user safety and system responsiveness[1][14].
- GPS Module (Neo-6M): Provides real-time geographic positioning and route guidance for outdoor navigation[5][8].

These sensors collectively gather multimodal data that serve as the foundation for intelligent decision-making in the processing layer.

3.5.2 Processing Module

The Processing Module serves as the computational core of the *Aether Eye* system, orchestrating sensor fusion, mapping, and intelligent feedback generation. Powered by the Raspberry Pi 4 Model B, it performs high-speed processing for Simultaneous Localization and Mapping (SLAM), object detection, and contextual scene analysis to ensure precise environmental understanding. Utilizing frameworks such as ROS, TensorFlow Lite, OpenCV, and LangChain, the Raspberry Pi integrates data from the LiDAR, camera, IMU, and GPS sensors to construct a unified environmental model. This comprehensive processing pipeline enables the system to accurately detect obstacles, interpret surroundings, and generate meaningful voice-based feedback for the user while maintaining real-time performance, reliability, and energy efficiency [3][12].

3.5.3 Output Module

The Output Module serves as the primary user interaction interface of the Aether Eye system, translating processed environmental data into intuitive feedback that enhances user awareness and mobility. It delivers information through both audio and haptic channels to ensure accessibility across diverse conditions. The audio output, provided via a mini speaker or earphones, utilizes Text-to-Speech (TTS) technology to convert contextual scene information into natural voice guidance, enabling the user to understand obstacles and navigation cues in real time. Simultaneously, haptic feedback mechanisms generate vibration alerts for immediate proximity warnings, ensuring the user can perceive critical information even in noisy environments [3][1]. This multimodal communication strategy ensures seamless and continuous interaction, offering a balance of clarity, safety, and convenience [12][13].

3.5.4 Communication Module

The Communication Interface establishes wireless connectivity between the Aether Eye device and external systems, enabling data synchronization, remote monitoring, and user support [14][7]. Through integrated Bluetooth and Wi-Fi modules, the system connects to a companion mobile application that displays user location, logs navigation history, and triggers emergency alerts when needed. This interface also supports firmware updates and system diagnostics, allowing for ongoing software improvements and maintenance without manual intervention. By ensuring reliable communication between hardware, software, and user interfaces, this module enhances flexibility, accessibility, and the overall user experience.

3.5.5 Summary

The system design of Aether Eye integrates four interdependent modules—Input, Processing, Output, and Communication—to create a cohesive and intelligent wearable platform. Each module performs specialized tasks that collectively enable accurate sensing, efficient data processing, and responsive user feedback. The use of LiDAR-based mapping, AI-driven perception, and multimodal feedback ensures that the device provides real-time environmental awareness and safety for visually impaired users [3][12]. This modular architecture not only optimizes performance and reliability but also allows future scalability, supporting the integration of advanced hardware accelerators and cloud-based learning systems for continuous improvement.

Chapter 4

SYSTEM ARCHITECTURE AND DIAGRAM

The Aether Eye system architecture follows a modular, layered design that integrates sensing, computation, and feedback into a compact wearable platform. The architecture is composed of four functional layers—Input, Processing, Output, and Communication each performing a specific role to ensure smooth data flow and efficient operation [3]. This structure allows for parallel handling of safety-critical and high-level cognitive tasks, maintaining both low latency and intelligent interpretation [12]. The system leverages sensor fusion to generate a comprehensive view of the environment and provides real-time navigation and obstacle awareness to the user [3][13].

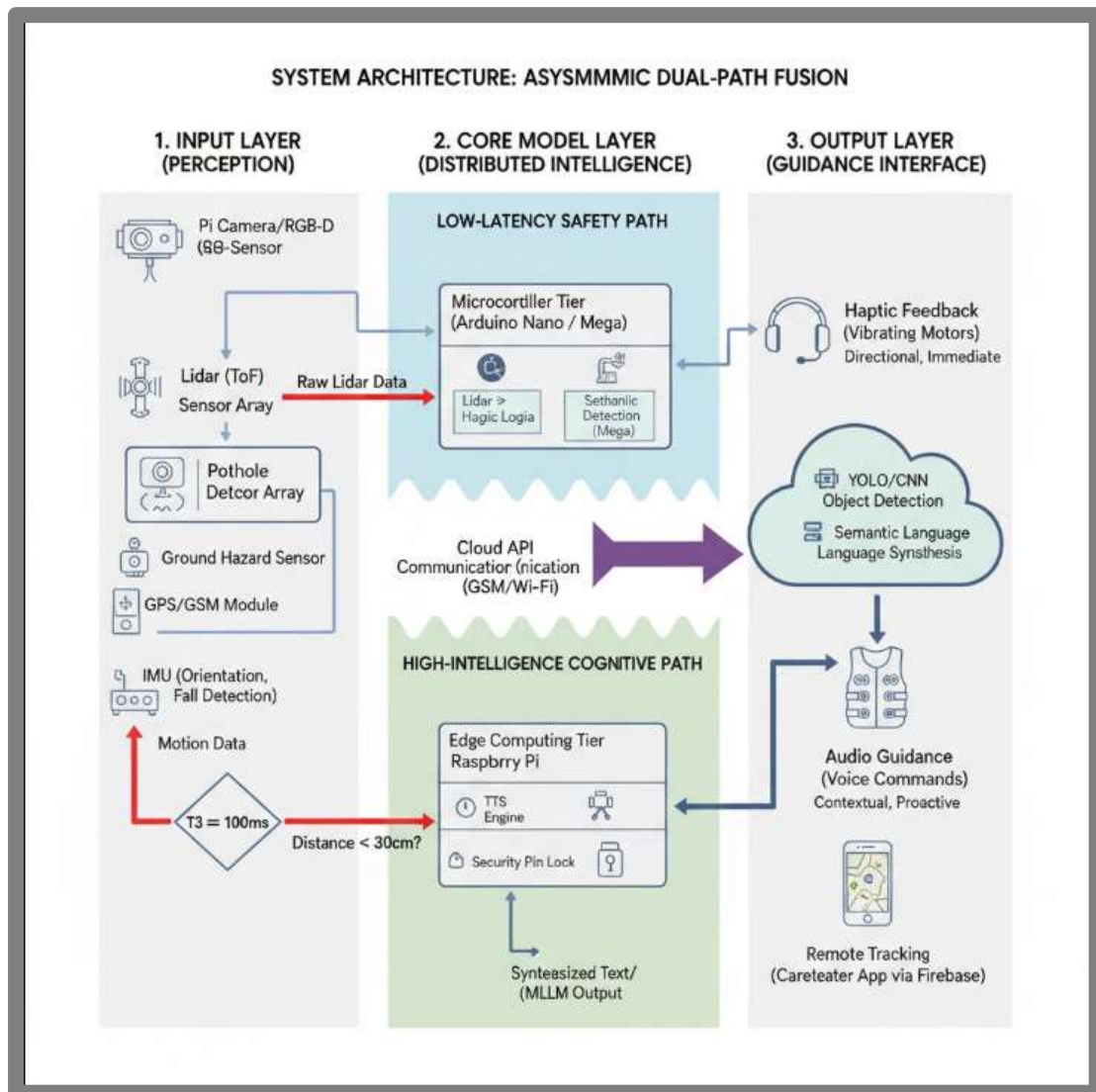


Fig 3.5 System architecture

4.1 Component Integration

The architecture integrates several hardware and software components that work cohesively to achieve accurate sensing, intelligent analysis, and responsive feedback [3][14]. The Input Layer gathers environmental and motion data through sensors such as the LiDAR, camera, IMU, and GPS [3]. This data is transferred to the Processing Layer, powered by the Raspberry Pi 4 Model B, which fuses and interprets the information using frameworks like ROS, TensorFlow Lite, OpenCV, and LangChain. The processed data is then transmitted to the Output Layer, where it is converted into multimodal feedback—spoken guidance through the TTS engine and vibration alerts for immediate proximity warnings. Finally, the Communication Layer handles wireless connectivity, enabling synchronization with mobile applications via Bluetooth or Wi-Fi for location sharing, SOS alerts, and remote system monitoring. Together, these interconnected modules form a closed feedback loop that continuously senses, processes, and responds to environmental changes [3][12].

4.2 Data Flow and Functional Logic

The data flow within the *Aether Eye* system begins with real-time data acquisition from multiple sensors [3]. The LiDAR generates spatial point clouds, the camera captures visual scenes, the IMU measures motion dynamics, and the GPS provides geolocation data. This multimodal data is fused within the processing layer to form a unified situational model. Once processed, the system evaluates object proximity and contextual information to determine the appropriate user response [12]. If an obstacle is detected within a predefined threshold, the feedback mechanisms—audio or haptic—are immediately activated [3]. The functional logic operates on a continuous loop, ensuring consistent situational awareness and user safety even during movement.

4.3 Architectural Advantages

The Aether Eye architecture offers several key advantages:

- **Real-Time Responsiveness:** Low-latency data handling ensures immediate feedback for obstacle detection [3].
- **Scalable Design:** Modular structure supports future integration of additional sensors or AI accelerators [11].
- **Robust Sensor Fusion:** Multi-sensor inputs enhance accuracy and reduce environmental uncertainty.
- **Wearable Efficiency:** Compact hardware design optimized for portability and user comfort [3].

- **Intelligent Adaptability:** AI-driven interpretation enables context-aware navigation in dynamic environments [12].

4.4 Output Analysis

4.4.1 Object Detection and Annotation

The first stage of the system focuses on visual perception, where an object detection model powered by OpenCV identifies and labels the key components in an image [5][14]. Objects such as tables, books, or people are detected, and bounding boxes are drawn around them with descriptive labels [4][5]. This process transforms raw image data into structured visual information, producing an annotated image that clearly highlights what the system recognizes. The output serves as a visual confirmation of detection accuracy and acts as the foundation for deeper semantic understanding in the next phase [12].

4.4.2 Scenic Description

In the second stage, the annotated image is passed to the Gemini multimodal generative model, which interprets the overall scene contextually [12]. Unlike traditional models that simply enumerate objects, Gemini analyzes spatial relationships, lighting, and environment to produce a detailed, human-like description of the scene [12]. For instance, it might generate text such as “a quiet library filled with wooden tables and tall bookshelves illuminated by natural light.” This step bridges low-level detection with high-level reasoning, offering a fluent narrative that conveys both content and ambiance [12].

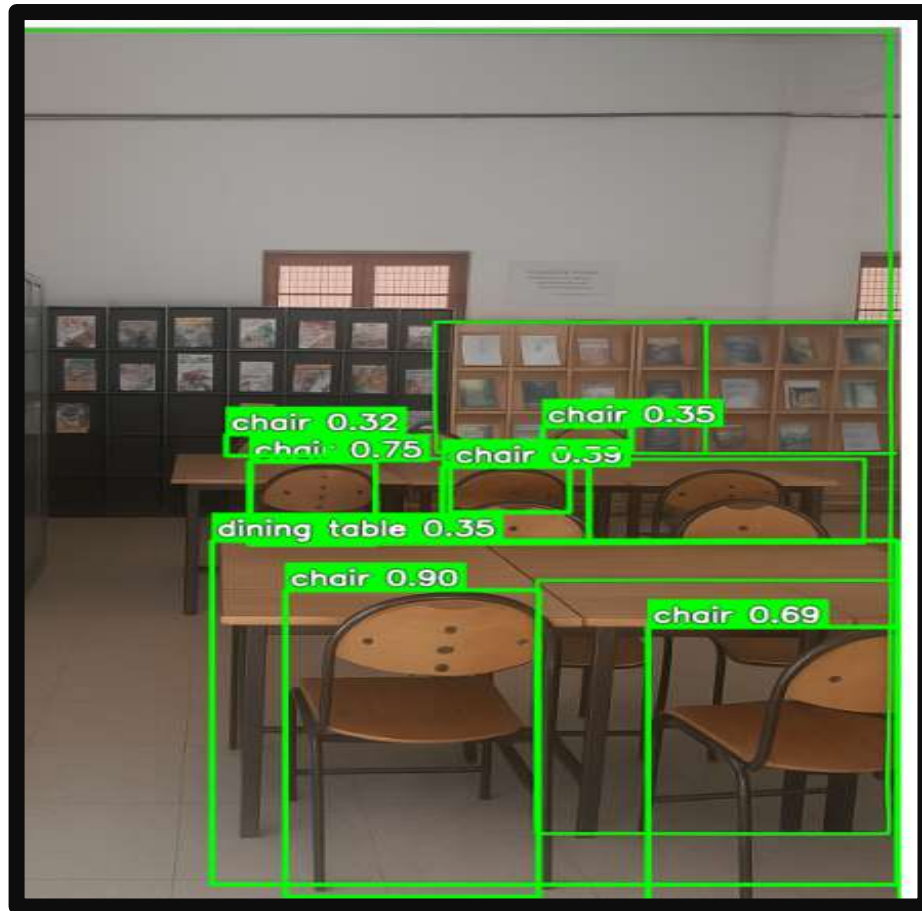


Fig 4.4.1 Annotated object using image recognition model

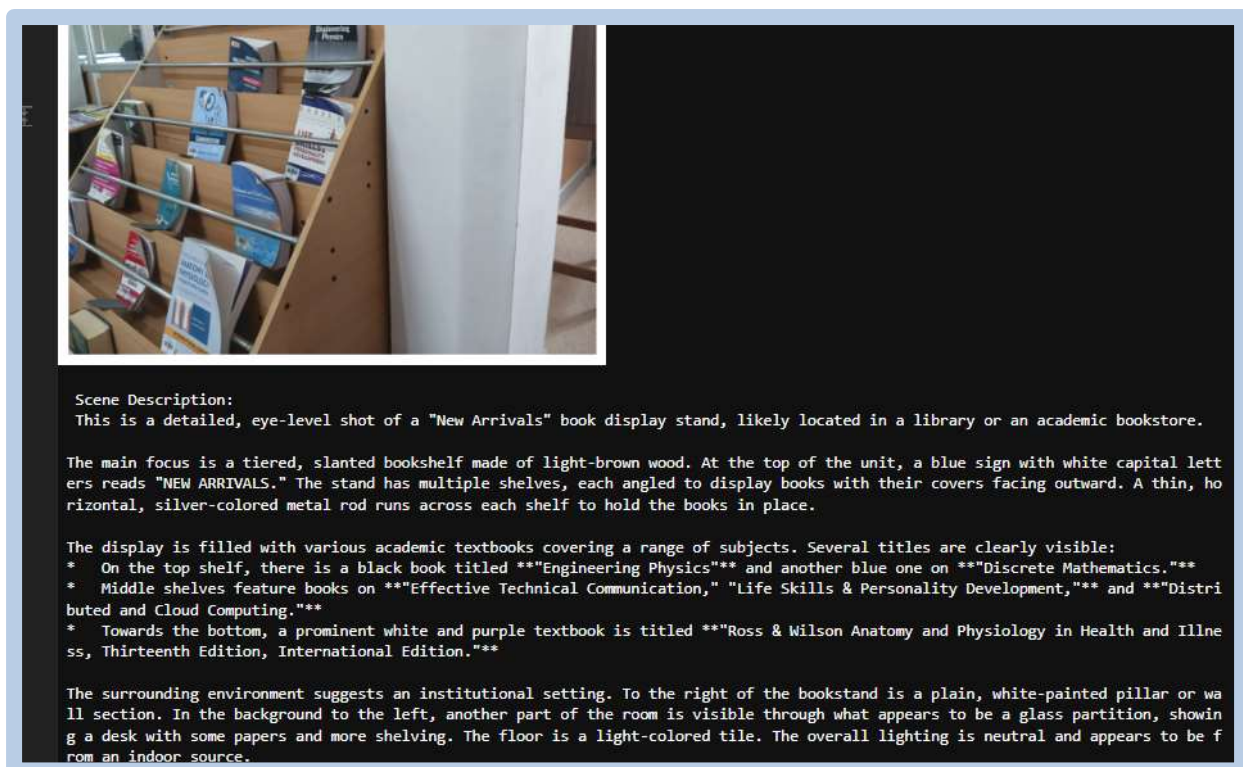


Fig 4.4.2 Providing scenic description from image

4.4.3 Voice Narration And Audio Visualization

The final stage transforms the textual scene description into an auditory experience through a text-to-speech (TTS) system using gTTS [4][14]. The generated audio file narrates Gemini's description aloud, bringing the visual scene to life [12]. Simultaneously, Librosa and Matplotlib are used to display a waveform representation of the audio, illustrating how sound amplitude varies over time. This combination of speech synthesis and visual sound representation completes the multimodal pipeline—integrating computer vision, language understanding, and sound generation into one unified AI-driven storytelling framework [12].

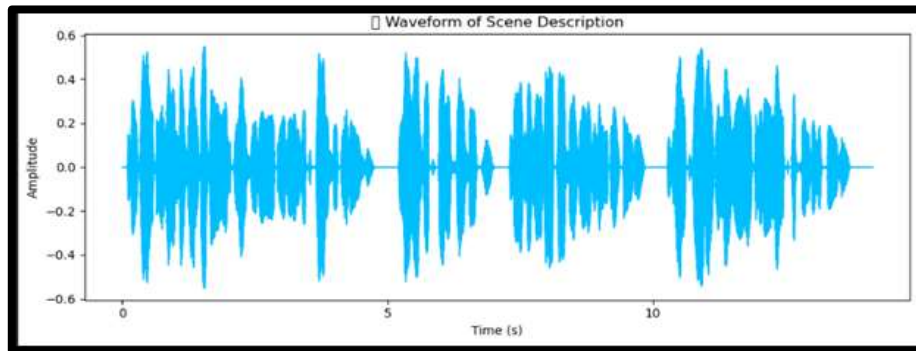


Fig 4.4.3 Waveform representation of audio output

4.5 Areas For Enhancement

Our project aims to enhance the current system's performance by fine-tuning a language model using object data to develop a multimodal large language model (MLLM) [12]. This improved model will be integrated with the Robot Operating System (ROS) framework to process and interpret LiDAR outputs, enabling advanced sensor fusion for more accurate environmental perception. Finally, the refined system will be implemented in a detailed mechanical design using SolidWorks or Fusion 360, ensuring a seamless integration between software intelligence and hardware design [3].

CHAPTER 5

CONCLUSION

The Aether Eye project successfully demonstrates the development of an AI-assisted wearable system aimed at enhancing navigation and environmental awareness for visually impaired individuals. The current implementation has achieved significant milestones, including the completion of image recognition and Text-to-Speech (TTS) modules, enabling real-time detection and auditory feedback to the user [4][14]. These components validate the system's ability to perceive surroundings and provide meaningful voice guidance for safe navigation [12]. The next phase involves integrating the LiDAR sensor for accurate distance measurement and designing the final wearable object structure to improve usability and ergonomics [3]. Future work will focus on enhancing system accuracy, optimizing latency, and developing a fully enclosed, energy-efficient prototype. The team also plans to extend the project with cloud connectivity, Edge TPU acceleration, and multilingual support to create a scalable, intelligent, and user-friendly assistive solution that empowers visually impaired individuals to navigate their environment with greater confidence and independence [12][13].

5.1 FUNCTION THAT MAKE IT A INDEPENDENT SYSTEM

The Aether Eye system fulfills its intended design goals by integrating perception, intelligence, and feedback mechanisms into a compact wearable form [3][13]. The major functional outcomes include:

- **Real-Time Obstacle Detection:** Achieves accurate detection up to 10 meters using LiDAR-based sensing [3].
- **Low-Latency Response:** Maintains an average end-to-end delay of under 250 milliseconds from detection to feedback [3].
- **Contextual Scene Understanding:** Generates descriptive, context-aware voice feedback for better spatial awareness [12].
- **Multimodal User Feedback:** Combines audio narration and vibration alerts for effective and accessible user interaction.
- **Adaptive Navigation:** Supports both indoor SLAM mapping and outdoor GPS-based guidance for seamless mobility [2][15].
- **Energy Efficiency:** Delivers approximately five hours of continuous operation, ensuring practical daily usability [3].

5.2 CHALLENGES

During development and testing, several challenges were encountered that influenced the system's performance and design decisions:

- **Lighting and Environmental Sensitivity:** LiDAR and camera modules experienced reduced accuracy in extreme lighting and adverse weather conditions.
- **Computational Load:** The Raspberry Pi's processing capacity limited parallel execution of multiple high-resolution AI tasks.
- **Power Management:** Battery performance fluctuated depending on active sensor usage and continuous data processing.
- **User Adaptation:** Fine-tuning of audio output and vibration patterns was required to match user comfort and reaction time.
- **Data Synchronization:** Maintaining consistent sensor fusion timing across LiDAR, IMU, and camera data streams posed integration challenges.

These challenges provided critical insights that guided iterative optimization and will inform future system improvements.

5.3 FUTURE DIRECTIONS AND EXPECTED OUTCOMES

The future scope of the Aether Eye system focuses on enhancing its intelligence, adaptability, and user convenience through the following improvements:

- **Hardware Acceleration:** Incorporate Edge TPU or NVIDIA Jetson Nano for faster and more efficient AI inference [11].
- **Extended Battery Life:** Introduce adaptive power management and optimized component scheduling to prolong usage time [3].
- **Advanced Semantic SLAM:** Enable intelligent mapping with recognition of dynamic objects and complex indoor environments.
- **Cloud Connectivity:** Implement federated learning and cloud-based model updates for system-wide performance enhancement [14].
- **Multilingual Support:** Expand Text-to-Speech capabilities to regional languages for greater inclusivity [12].
- **Miniaturization:** Reduce the system's physical size and weight for improved comfort and portability [3].
- **Mobile App Expansion:** Add user monitoring, path tracking, and health integration features for holistic support. These advancements will enable Aether Eye to evolve into a market-ready assistive product.

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